2022 SSMO Tiebreaker Round Solutions

SMO Team

Tiebreaker Round Problem 1: For all positive integers n, let $S(n)$ denote the least positive integer x such that $n + x$ is a palindrome. Find the value of $\sum_{n=1}^{100} S(n)$.

Answer: $\boxed{542}$

Solution: Let $P(n)$ denote $n+S(n)$. For $n = 1, \ldots, 8$, we have $P(n) = n+1$. For $n = 9, 10$, we have $P(n) = 11$. For each subsequent block of size 11, we have $P(n) = 22, 33, \ldots, 99$. Finally, for $n = 99, 100$, we have $P(n) = 101$. So,

$$\begin{aligned}\sum_{n=1}^{100} S(n) &= \sum_{n=1}^{100} P(n) - \sum_{n=1}^{100} n \\ &= (2 + \cdots + 9) + 2 \cdot 11 + 11 \cdot (22 + \cdots + 99) + 2 \cdot 101 - 5050 \\ &= 44 + 22 + 5324 + 202 - 5050 = \boxed{542}.\end{aligned}$$

Tiebreaker Round Problem 2: Let $P(x) = x^3 - 7x^2 + 9x - 2$. If $P(x) = (x-a)^3 + b(x-a) + c$ where a, b, c are real numbers, then the value of $a - b - c$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{461}$

Solution: Expanding, we have

$$P(x) = x^3 - (3a)x^2 + (3a^2 + b)x - (a^3 + ab - c).$$

Comparing the coefficients of x^2, we have $3a = 7 \implies a = \frac{7}{3}$. Comparing the coefficients of x, we have

$$3a^2 + b = 9 \implies 3\left(\frac{7}{3}\right)^2 + b = 9 \implies b = -\frac{22}{3}.$$

Finally, comparing the constant terms, we have

$$a^3 + ab - c = 2 \implies \left(\frac{7}{3}\right)^3 + \left(\frac{7}{3}\right)\left(-\frac{22}{3}\right) - c = 2 \implies c = -\frac{173}{27}.$$

So,

$$a - b - c = \frac{7}{3} - \left(-\frac{22}{3}\right) - \left(-\frac{173}{27}\right) = \frac{434}{27} \implies 434 + 27 = \boxed{461}.$$

Tiebreaker Round Problem 3: Find the sum of the 5 smallest positive prime numbers p such that $3^n + 4^n \equiv 0 \pmod{p}$ has no positive integer solutions for n.

Answer: $\boxed{76}$

Solution: Note that there are no solutions when $p = 2, 3$. Assume otherwise. We have

$$3^n + 4^n \equiv 0 \implies 3^n \equiv -4^n \implies \left(3 \cdot 4^{-1}\right)^n \equiv -1 \pmod{p}.$$

Let $a = 3 \cdot 4^{-1}$. From Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod{p}$. The equation has no solutions if and only if $a^d \equiv 1 \pmod{p}$, where $d = \frac{p-1}{2^{v_2(p-1)}}$ is the largest odd integer dividing $p-1$. Consider the following table

p	$a = 3 \cdot 4^{-1} \pmod{p}$	$d = \frac{p-1}{2^{v_2(p-1)}}$	$a^d \pmod{p}$
5	2	1	$2^1 \equiv 2 \pmod{5}$
7	6	3	$6^3 \equiv 6 \pmod{7}$
11	9	5	$9^5 \equiv 1 \pmod{11}$
13	4	3	$4^3 \equiv 12 \pmod{13}$
17	5	1	$5^1 \equiv 5 \pmod{17}$
19	15	9	$15^9 \equiv 18 \pmod{19}$
23	18	11	$18^{11} \equiv 1 \pmod{23}$
29	8	7	$8^7 \equiv 17 \pmod{29}$
31	24	15	$24^{15} \equiv 30 \pmod{31}$
37	10	9	$10^9 \equiv 1 \pmod{37}$

So, our answer is

$$2 + 3 + 11 + 23 + 37 = \boxed{76}.$$